

Build Report — Build mileage tracker app

Chain 20260610T015817Z_3289dd · Story story_8fbb58 · Repo raghu-cmyk/mileage-tracker-v6 @ feat/chain-20260610T015817Z
Generated 20260610T022030Z

1. Job Summary (plan → implement → validate → deliver → report)

Order	Job ID	Type	State	Duration
1	20260610T015817Z_277159	plan	completed	1m 6s
2	20260610T015924Z_0ed8db	implement	completed	3m 44s
3	20260610T020309Z_09605f	validate	completed	14m 40s
4	20260610T021749Z_a6cafd	deliver	completed	1m 17s
5	20260610T021906Z_fcd192	report	running	0s

2. Requirements → GitHub Issues Mapping

Issue	Title	Requirements Covered
#17	Implementation Approach	Product architecture; §1 overview, stack selection
#18	Authentication & Session Management	§2 Auth (FR-AUTH1–4), NFR-4 security
#19	Vehicle & Odometer Management	§3 Vehicles (FR-V1–4), annual odometer readings
#20	IRC §274(d) Trip Management with Audit Trail	§4 Trips (FR-T1–6), audit trail, late-entry
#21	Time-Effective IRS Mileage Rates & Trip Categories	§5 Rates (FR-R1–3), DB reference data
#22	Deduction Calculation & Year-End Summary	§6 Deductions (FR-D1–3), business-use %
#23	CSV & PDF Export	§7 Exports (FR-E1–2), IRS substantiation columns
#24	Receipt Capture (Upload, Storage, Display & Export)	§9 Receipt Capture (FR-RC1–5)
#25	Design & Aesthetic — Visual Design System	§8 UI (polished tokens, dark mode, responsive)

3. Per-Job Agent Activity Log (excerpts)

```
plan — 20260610T015817Z_277159
## Planning complete

Read `docs/requirements/mileage-tracker-requirements.md` from raghu-cmyk/mileage-tracker-v6 (via
GitHub). The target repo already has a complete planning backlog from the prior v6 chain run; those issues
match the current requirements, so no duplicates were created. The `gate_min_issues` gate passes (24 issues ≥
3).

### Chosen implementation approach

Documented in [#17 – Implementation Approach](https://github.com/raghu-cmyk/mileage-tracker-v6/issues/17):

| Candidate | Summary | Key trade-off |
|-----|-----|-----|
| **1. Next.js monolith** (chosen) | App Router + PostgreSQL + Prisma + blob storage | Best
UI/dark-mode/responsive path; blob storage is explicit |
| 2. Django + HTMX | Server-rendered Python monolith | Strong auth/audit, weaker polished SPA-like UI |
| 3. React SPA + FastAPI | Decoupled frontend + API | Clean separation, but two deploys and auth complexity
for single-user scope |
```

****Winner:**** Next.js full-stack monolith – \$8 UI requirements, NFR-4 server-side auth, exact decimal math, and single-deploy simplicity for a single-user app.

Feature issues (`plan-task`)

Issue	Title
[#18](https://github.com/raghu-cmyk/mileage-tracker-v6/issues/18)	Authentication & Session Management
[#19](https://github.com/raghu-cmyk/mileage-tracker-v6/issues/19)	Vehicle & Odometer Management
[#20](https://github.com/raghu-cmyk/mileage-tracker-v6/issues/20)	IRC §274(d) Trip Management with Audit Trail
[#21](https://github.com/raghu-cmyk/mileage-tracker-v6/issues/21)	Time-Effective IRS Mileage Rates & Trip Categories
[#22](https://github.com/raghu-cmyk/mileage-tracker-v6/issues/22)	Deduction Calculation & Year-End Summary
[#23](https://github.com/raghu-cmyk/mileage-tracker-v6/issues/23)	CSV & PDF Export
[#24](https://github.com/raghu-cmyk/mileage-tracker-v6/issues/24)	Receipt Capture (Upload, Storage, Display & Export)

Design issue (`design`)

Issue	Title
...	(11 more lines truncated)

implement — 20260610T015924Z_0ed8db

****Implement stage complete**** for chain `20260610T015817Z\_3289dd` / story `story\_8fbb58` ("Build mileage tracker app").

What was delivered

Implemented ****Mileage Tracker v6**** (Next.js 14 monolith per plan issue #17) covering plan tasks #18–#25:

- Authentication (Argon2id sessions, rate limiting)
- Vehicle & odometer management
- IRC §274(d) trip logging with audit trail
- Time-effective IRS mileage rates
- Deduction calculation & year-end summary
- CSV/PDF export
- Receipt capture
- Design system (Ledger palette, dark mode, responsive layout)

Verification

Check	Result
`npm test`	15 passed
`npm run smoke`	PASS (register, login, vehicle, trip – no 5xx)
`no_todo_markers` gate	Clean

Output artifacts

- ****Job workspace:**** `var/projects/intent-network-mission-control/var/workspaces/20260610T015924Z\_0ed8db/`
- ****Stage output:**** `artifacts/output.json`
- ****GitHub branch:**** `feat/chain-20260610T015817Z` on `raghu-cmyk/mileage-tracker-v6`
- ****Commit:**** `030017614565f5bde787235e67808152fdaf005`

MC reporting

...
(6 more lines truncated)

validate — 20260610T020309Z_09605f

Validation ****passed**** for the mileage tracker app on `raghu-cmyk/mileage-tracker-v6` @ `feat/chain-20260610T015817Z`.

All key GET pages returned 2xx, and the primary POST flows succeeded with no 5xx:

- `/api/auth/register` → 201
- `/api/auth/login` → 200
- `/api/vehicles` → 201
- `/api/trips` → 201

Evidence is in the job workspace as `validation\_evidence.json` and `VALIDATION\_PASSED`. The validate job reported success via `mc done`; the daemon should advance the `full-app-delivery` chain to the **deliver** stage next.

deliver — 20260610T021749Z_a6cafd

Delivery complete

Branch confirmed

- **Repository:** `raghu-cmyk/mileage-tracker-v6`
- **Head:** `feat/chain-20260610T015817Z` (1 commit ahead of `mc\_webhook\_test`)
- **Validated work:** Full Next.js mileage tracker MVP (auth, vehicles, trips, IRS rates, deductions, exports, receipts, design system)

Validation evidence

The validate stage passed on 2026-06-10:

- All key GET pages returned **200**
- POST flows: register **201**, login **200**, create vehicle **201**, create trip **201**
- **No 5xx** responses

Pull request opened

https://github.com/raghu-cmyk/mileage-tracker-v6/pull/28

- **Base:** `mc\_webhook\_test`
- **Title:** Build mileage tracker app – full MVP
- **Issues referenced:** Closes #17–#25
- **PR open gate:** Passed (`open\_prs=1`)

Job signaled

`mc done --summary "Opened PR for Build mileage tracker app into mc\_webhook\_test"` recorded for job `20260610T021749Z\_a6cafd`. The chain supervisor will advance to the **report** stage next.

report — 20260610T021906Z_fcd192

4. Validation Results (Gate Outcomes)

Stage	Gate	Outcome
plan	gate_min_issues.sh 3	PASSED — 24 planning issues on target repo (≥3 required)
implement	no_todo_markers (workspace)	PASSED — no placeholder markers; npm test 15/15; smoke OK
validate	test -f VALIDATION_PASSED	PASSED — all GET 200; POST register=201, login=200, vehicles=201, trips=201; no 5xx
deliver	gate_pr_open.sh mc_webhook_test	PASSED — PR open: https://github.com/raghu-cmyk/mileage-tracker-v6/pull/28
report	find docs/build-reports/report-*.pdf -size +2M	PENDING — this report

VALIDATION_PASSED: All GET pages loaded; POST flows OK (/api/auth/register=201, /api/auth/login=200, /api/vehicles=201, /api/trips=201); no 5xx

5. Total Build Time

20m 48s (01:58:18Z → 02:19:06Z UTC) (plan through deliver; report in progress)

6. Application Code Stats

Metric	Value
Source files (excl. node_modules, .next)	64
Lines of code (TS/TSX/JS/CSS/Prisma)	4842
Unit tests passing	15/15

Delivered branch	feat/chain-20260610T015817Z
Pull request	https://github.com/raghu-cmyk/mileage-tracker-v6/pull/28